AI EXPLAINED SIMPLY

The AI Chip Map: Why Almost Every AI Chip Is Made in Taiwan

A one-page map of where an advanced AI chip comes from, with every number sourced. Companion to the AI Explained Simply video.

The chain in 4 steps

1. **Design (USA)**: Nvidia designs Blackwell in Santa Clara, California. Nvidia owns no factory that makes it ("fabless").
2. **Machine (the Netherlands)**: ASML, in Veldhoven, is the only company that sells EUV lithography machines. A laser hits drops of molten tin up to 50,000 times per second to make 13.5 nm light. Reuters puts one machine at about $250 million.
3. **Make (Taiwan)**: TSMC prints the chip on silicon wafers. Blackwell uses TSMC's 4NP process. Advanced fabs sit in Hsinchu (north) and Tainan (south, Fab 18: 5 nm and 3 nm).
4. **Package (Taiwan, for now)**: each chip is joined to its memory with TSMC's CoWoS advanced packaging. In 2025, wafers made in Arizona still flew back to Taiwan for this step.

Key numbers

- **69.9%**: TSMC's share of global foundry revenue in 2025 (TrendForce), $122.5 billion in sales. Samsung was second with 7.2%.
- **58%**: share of TSMC's 2025 revenue from high-performance computing, mostly AI.
- **92%**: Taiwan's share of the world's most advanced (below 10 nm) chipmaking capacity in 2021 (BCG x SIA).
- **58,000 m²**: clean room in each phase of Fab 18, about double a standard logic fab. Total investment: more than NT$1.86 trillion.
- **$165 billion**: TSMC's total planned investment in Arizona (March 2025): 6 fabs, 2 advanced packaging plants, 1 R&D centre.

Timeline

- **Dec 15, 1980**: Hsinchu Science Park opens.
- **1985**: Morris Chang (25 years at Texas Instruments) is asked to lead Taiwan's ITRI.
- **Feb 21, 1987**: TSMC is founded in Hsinchu, the first "pure-play" foundry: it makes chips for others and never competes with its customers.
- **1993**: Nvidia is founded, without a factory.
- **Dec 29, 2022**: 3 nm volume production starts at Fab 18, Tainan.
- **Feb 24, 2024**: TSMC opens its first Japanese fab in Kumamoto (partners: Sony, Denso, Toyota), for older chips, not AI.
- **Apr 3, 2024**: strongest quake in Taiwan in about 25 years. More than 70% of TSMC's tools were running again within 10 hours.
- **Jul 2025**: Rapidus (Japan) shows its first 2 nm test chips in Chitose, Hokkaido. Mass production goal: 2027.
- **Oct 2025**: first US-made Nvidia Blackwell wafer at TSMC Arizona.

3 things to remember

1. **A business model**: make chips for everyone, compete with no one. So every chip designer came.
2. **A cluster**: engineers, suppliers and repair teams within an hour's drive. One factory abroad cannot copy that.
3. **Scale**: 70% of the market means more practice and fewer mistakes than anyone else.

Glossary

- **Foundry**: a factory that makes chips designed by other companies.
- **Fabless**: a chip company that designs but does not manufacture (Nvidia, AMD, Apple).
- **EUV lithography**: printing a chip's pattern with extreme ultraviolet light (13.5 nm).
- **Advanced packaging (CoWoS)**: joining the chip and its memory so they work as one AI chip.
- **nm (nanometre)**: a billionth of a metre; the name of a process generation (5 nm, 3 nm, 2 nm).

Sources

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